

Practicals: User Activity Keystroke Analysis

1. Introduction:

This practical involves using Weka Explorer to analyze user activity through keystroke dynamics. The goal is to classify users based on their typing behavior using two machine learning algorithms: Random Forest and Multilayer Perceptron(MLP). We trained both classifier and compared their performance in terms of accuracy, Kappa statistic, and other metrics.

2. Model Comparison:

Metric	Random Forest	MLP	Difference	Winner	Comment
Accuracy (%)	89.60	89.60	0	Tie	Both models performed equally well.
Kappa Statistic	0.8939	0.8939	0	Tie	Same agreement score.
Precision	0.88	0.898	0	Tie	Identical weighted precision.
Recall	0.896	0.896	0	Tie	Same recall value.
F1 Score	0.896	0.896	0	Tie	No difference.
Training Time (s)	63.21	Longer	MLP Slower	Random Forest	Random Forest is more efficient

3. Analysis & Discussion

Both Random Forest and Multilayer Perceptron achieved identical results on the keystroke dataset, with an accuracy of 89.6% and Kappa statistic of 0.8939. This indicates that both models are capable of capturing user behavior patterns accurately. However, MLP generally requires more training time due to its neural network structure.

Random Forest, being an ensemble of decision trees, is faster to train and easier to interpret. Thus, if computational efficiency is important, Random Forest is the preferred choice. In cases where deeper feature learning is needed, MLP can be considered.

4. Screenshots:

The Random Forest

```
=== Summary ===

Correctly Classified Instances      1828      89.6078 %
Incorrectly Classified Instances    212      10.3922 %
Kappa statistic                     0.8939
Mean absolute error                 0.0113
Root mean squared error             0.0634
Relative absolute error             29.279 %
Root relative squared error         45.7344 %
Total Number of Instances          2040

=== Detailed Accuracy By Class ===
```

MLP

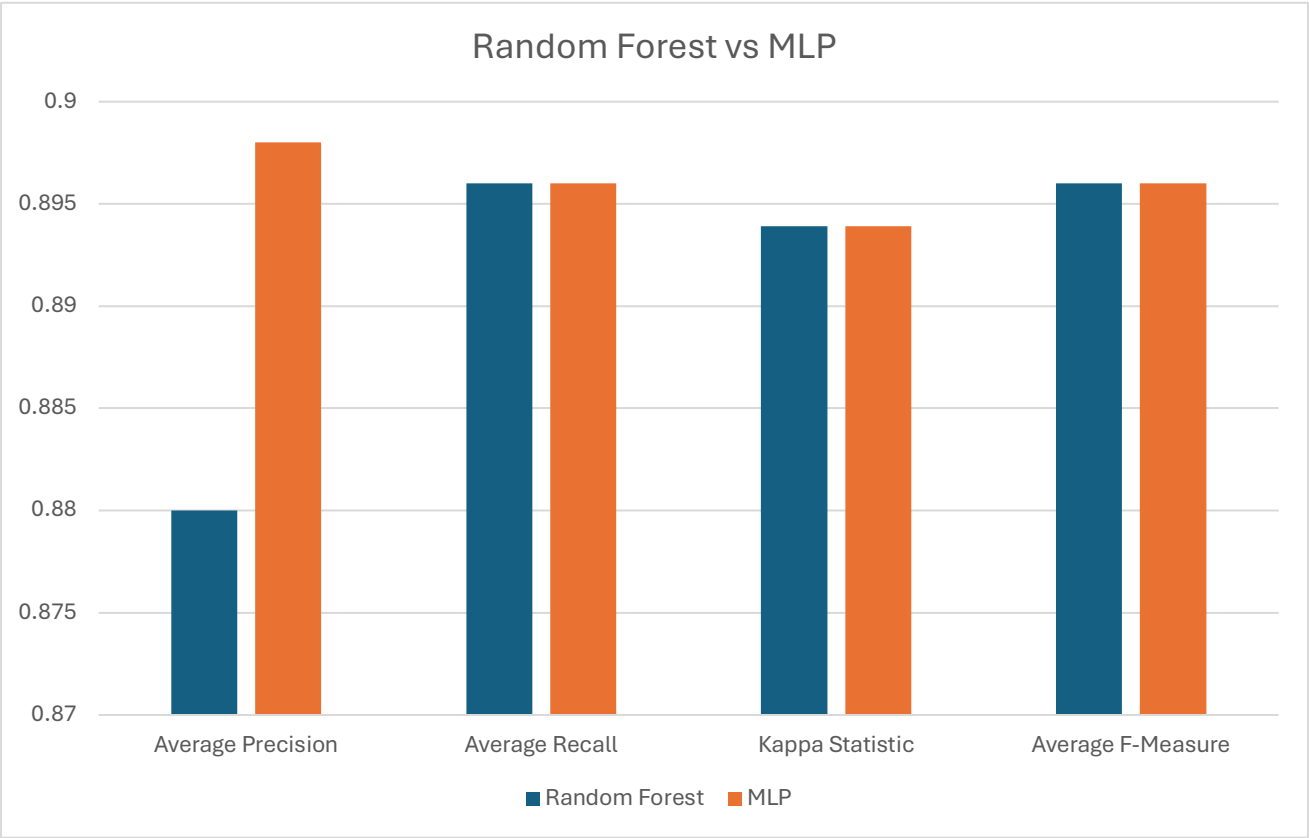
```
=== Summary ===

Correctly Classified Instances      37      1.8137 %
Incorrectly Classified Instances    2003     98.1863 %
Kappa statistic                     0
Mean absolute error                 0.0385
Root mean squared error             0.1388
Relative absolute error             100.0096 %
Root relative squared error         100.1074 %
Total Number of Instances          2040
```

Comparative Analysis: MLP vs Random Forest

Table 2: Summary of results of Classification Model

Model	Average Precision	Average Recall	Kappa Statistic	Average FMeasure
Random Forest	0.896	0.896%	0.8939	0.896
Multilayer Perceptron (MLP)	0.896	0.896	0.8939	0.896



Conclusion

This practical demonstrated how machine learning models can be applied to behavioral biometric data such as keystrokes. With proper preprocessing and model selection, we can effectively classify users based on their typing patterns. Both Random Forest and MLP showed strong performance, validating the reliability of the dataset and the methods used.